MEASURING AN EQUILIBRIUM CONSTANT

OBJECTIVE

- To determine K_c for the ion formed by the reaction of Fe^{3+} and SCN^- using Le Chatelier's Principle and a Beer's Law Calibration plot.

INTRODUCTION

A traditional test for the presence of the ferric ion (Fe^{3+}) in aqueous solutions is to add thiocyanate (SCN^-), which produces a wine-red color that is indicative of iron(III) thiocyanate, ($FeSCN^{2+}$). While other metal cations react with SCN^-, only Fe^{3+} produces this particular color, so the test is very selective. Furthermore, the coloration is very intense, making the test sensitive. The reaction is simply:

$$Fe^{3+}_{(aq)} + SCN^-_{(aq)} \leftrightarrow FeSCN^{2+}_{(aq)}$$

And its corresponding equilibrium constant expression is

$$K_c = \frac{[FeSCN^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^-]_{eq}} \quad (1)$$

where $[Fe^{3+}]_{eq}$ is the concentration of Fe^{3+} at equilibrium, etc. In this lab you will accurately determine K_c for this reaction.

In principle, $[FeSCN^{2+}]_{eq}$ can be measured using spectroscopy and Beer's Law, based on the absorption of $[FeSCN^{2+}]$ at 450 nm (A_{450}). Knowing the initial concentrations of Fe^{3+} (i.e. $[Fe^{3+}]_{in}$), SCN^- (i.e. $[SCN^-]_{in}$) and $[FeSCN^{2+}]_{eq}$, we can calculate K_c using Table 1.

Table 1: Theoretical ICE Table and reaction

$$Fe^{3+}_{(aq)} \quad + \quad SCN^-_{(aq)} \quad \leftrightarrow \quad FeSCN^{2+}_{(aq)}$$

I	Initial $[Fe^{3+}]$	Initial $[SCN^-]$	0
C	$-x$	$-x$	$+x$
E	Initial $[Fe^{3+}] - x$	Initial $[SCN^-] - x$	$x = [FeSCN^{2+}]_{eq}$

The challenge is that in order to prepare a Beer's Law calibration graph for $FeSCN^{2+}$ we need to know its equilibrium concentration (i.e. $[FeSCN^{2+}]_{eq}$). To know $[FeSCN^{2+}]_{eq}$ in a set of "standard" (or calibration) solutions, you will use Le Chatelier's Principle to "push" the reaction in the forward direction. This will be accomplished by having $[SCN^-]_{in} \lll [Fe^{3+}]_{in}$ so that essentially all of the SCN^- in the solution is bound by Fe^{3+} (see example in Table 2). This allows you to assume that $[FeSCN^{2+}]_{eq} = [SCN^-]_{in}$. Once you generate a Beer's Law calibration plot for solutions with known $[FeSCN^{2+}]_{eq}$, you will be able to measure the concentration of $[FeSCN^{2+}]_{eq}$ in any solution!

Table 2: Example ICE Table

$$Fe^{3+}_{(aq)} \quad + \quad SCN^{-}_{(aq)} \quad \leftrightarrow \quad FeSCN^{2+}_{(aq)}$$

I	0.05	3.33×10^{-5}	0
C	-3.33×10^{-5}	-3.33×10^{-5}	$+3.33 \times 10^{-5}$
E	~0.05	"0"	3.33×10^{-5}

To accurately determine K_c, it is best to use comparable initial concentrations of Fe^{3+} and SCN^{-}, that is $[Fe^{3+}]_{in} \approx [SCN^{-}]_{in}$. In the second part of the experiment, you will prepare three "samples", each with varying $[FeSCN^{2+}]_{eq}$. Here, $[FeSCN^{2+}]_{eq}$will be determined by measuring A_{450} of each sample and using your calibration curve. $[Fe^{3+}]_{eq}$ and $[Fe^{3+}]_{eq}$will then be determined by subtracting $[FeSCN^{2+}]_{eq}$ from the initial concentrations of each of these ions (as in Table 1). Once you have obtained $[FeSCN^{2+}]_{eq}$, $[Fe^{3+}]_{eq}$, and $[Fe^{3+}]_{eq}$, you can readily solve for K_c. Keep in mind, the "LeChatelier-pushed" calibration is an imperfect approach that produces systematic error in K_c measurements. So, we can expect some variation in the K_c values measured for the three samples even though, in principle, all of the solutions you generate in this experiment are governed by the same K_c value.

PROCEDURE

NOTE: Prepare all colorimetric solutions for Parts A and B before taking any absorbance measurements. Use tables in your notebook for data and show all calculations.

A. Instructions on Cleaning Pipettes

Do this procedure over a sink.

1. Fill your pipette one quarter full with distilled water.
2. Remove the pipettor from the pipette while tilting the pipette on its side. DO NOT ALLOW THE LIQUID TO TOUCH THE PIPETTOR.
3. Twirl the pipette so all inside surfaces are rinsed.
4. Empty the pipette into the sink.
5. Repeat this process twice with distilled water.
6. Obtain a 250 mL beaker to use as a waste container.
7. Repeat the rinsing process once with the solution you will be measuring. Empty the pipette into the waste beaker.
8. Repeat this procedure between each different solution if you are using the same pipette.

B. Preparation of Beer's Law Calibration Standards

1. Because you cannot pipette from stock bottles, fill three appropriately labeled beakers with:
   a. 50 mL of 0.100 $M\ Fe^{3+}$ solution
   b. 50 mL of 0.200 $mM\ SCN^-$ (thiocyanate) solution
   c. 50 mL of 0.5 $M\ HNO_3$ (nitric acid) solution
2. Use the solutions listed above (a, b, and c) to prepare the 4 solutions listed in Table 3. Prepare these solutions in 50 mL or 150 mL beakers. Use pipettes to measure all solutions.

Table 3: Beer's Law Calibration Solutions

Solution #	**0.100 $M\ Fe^{3+}$**	**0.200 $mM\ SCN^-$**	**0.5 $M\ HNO_3$**	$[FeSCN^{2+}]_{eq}$	**Absorbance**
Blank	10.0 mL	0.0 mL	20.0 mL		
1	10.0 mL	5.0 mL	15.0 mL		
2	10.0 mL	10.0 mL	10.0 mL		
3	10.0 mL	20.0 mL	0.0 mL		

3. Do the pipetting carefully, for K_c results will be no better than the Beer's Law calibration on which they are based.
4. Make sure each solution is well mixed (gentle swirling is adequate). The solution containing no SCN^- is the control (blank) solution. The blank solution should remain colorless. However, solutions 1-3 should be orange. CONTACT YOUR INSTRUCTOR IMMEDIATELY IF YOU DO NOT SEE AN ORANGE COLOR AFTER ADDING ALL THREE SOLUTIONS TOGETHER.
5. Use the spectrometer to prepare a Beer's Law calibration graph at 450 nm at which the absorbance is maximum. Use this wavelength for all measurements. There are directions (on page 5) provided by each instrument or follow the instructions at the end of this lab. For concentration, use the relation $[FeSCN^{2+}]_{eq} = [SCN^-]_{in}$. Don't forget to use your blank as a data point ($[FeSCN^{2+}]_{eq} = [SCN^-]_{in} = 0\ M$). This gives you 4 solutions of **known** concentration which you will later use to make your calibration plot.
6. **Check in with your lab instructor to verify that your calibration curve is good to move on to your K_c measurements.**

C. Measurement of K_c

1. In order to not pipette out of your Nalgene, pour about 15 mL of each Fe^{3+} stock solution in Table 4 in three separate labeled beakers. Using these stock solutions, prepare the three solutions (4, 5, 6) described in Table 4. As before, do all the measuring with pipettes, cleaning them appropriately. Use dilution calculations to fill in Table 4.

Table 4: Solutions for K_c Measurement

Solution #	Fe^{3+}	$[Fe^{3+}]_{in}$	$0.2\ mM\ SCN^-$	$[SCN^-]_{in}$
4	$10.0\ mL\ of\ 4.00\ mM$		$10.0\ mL$	
5	$10.0\ mL\ of\ 6.00\ mM$		$10.0\ mL$	
6	$10.0\ mL\ of\ 8.00\ mM$		$10.0\ mL$	

2. Proceed to getting a spectrometer and following the instructions at the end of this manual. In step 10, you will measure the absorbance and calculate the $[FeSCN^{2+}]_{eq}$ for each of the three **unknown** solutions that you can then fill into Table 5.

   Table 5: Solution $[FeSCN^{2+}]_{eq}$ Absorbance

Solution #	$[FeSCN^{2+}]_{eq}$	Absorbance
4		
5		
6		

3. Subtract $[FeSCN^{2+}]_{eq}$ from $[Fe^{3+}]_{in}$ and $[SCN^-]_{in}$ to find $[Fe^{3+}]_{eq}$ and $[SCN^-]_{eq}$, using the simple stoichiometry of the reaction (**Make an ICE table for each in order to fill in Table 6!**). Use the equilibrium concentrations to calculate K_c for solutions 4-6.

   Table 6: Solution K_c Values

Solution #	$[Fe^{3+}]_{eq}$	$[SCN^-]_{eq}$	$[FeSCN^{2+}]_{eq}$	K_c
4				
5				
6				

4. Calculate your average K_c.
5. Be sure to include a conclusion for this week's lab notebook upload. Some ideas to include are a) a discussion of your calibration curve – how good was it? What information were you able to get using it? b) interpreting your three individual K_C values as well as your average K_C, c) possible errors and lab techniques

WASTE AND CLEANUP

1. Rinse all pipettes with distilled water THREE times and return them to the proper tub.
2. Put all left over solutions in the Waste Container, clean your beakers with soap and water, and put empty solution storage bottles in the box by the cart.

EQUILIBRIM LAB – SPECTROPHOTOMETER SET-UP

1. Connect the spectrophotometer to your computer.
2. Open LoggerPro (type Logger Pro in the search bar). The instrument should automatically connect and will display a rainbow background on the graph.
3. Select **EXPERIMENT→ CALIBRATE → SPECTROPHOTOMER** and allow the spectrometer to warm up for 90 seconds.
4. Grab a cuvette and use the single cuvette for all measurements today. Also, be sure that the light can pass through your sample (no fingerprints, no debris, no liquid on the outside, no discoloration on the cuvette, etc.). Be sure to properly rinse your cuvette between measurements both with distilled water and the next solution that you will be measuring.
5. Place your blank solution (0 M concentration) in the spectrometer and select **Finish Calibration → OK**.
6. Select the Rainbow bar graph icon in the menu bar along the top.
7. Select Absorbance vs. Concentration and scroll down to select the $450\ nm$ checkbox (or the value nearest to $450\ nm$).
8. With your 0 M (blank) solution in the cuvette holder select collect. Select Keep and enter 0 M. DO NOT PRESS STOP!
9. Continue with samples 1, 2 and 3. Once the absorbance reading stabilizes, select Keep and enter the concentration in **units of molarity**.
10. After all four known samples have been measured, select **ANALYZE → Linear Fit**. Press Stop and a window displaying the slope (m) and y-intercept (b) will appear. Record these values within your $y = mx + b$ equation for use in Part C of the lab.
11. PART C: Measure samples 4, 5 and 6 by placing each sample in the cuvette holder and recording the absorbance reading once it stabilizes. There is no need to press Keep. The value you record equals y in your $y = mx + b$ equation. Solve for x to determine $[FeSCN^{2+}]_{eq}$.